

Mathematical Foundations of Political Methodology

Lecture 3

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Sequences and Limits

- We characterize the behavior of smooth functions with limits.
- To understand a limit, we need to have a unit of account, which is a sequence.

- Sequence
- Cauchy Sequence
- Limits of Functions
- Continuity

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Sequence

Sequence can be irregular, an example of such sequence is:

$$a_1 = 1$$

$$a_2 = 8$$

$$a_3 = 6$$

$$a_4 = \dots$$

Sequence can be regular, an example of such sequence is:

$$x_1 = 2$$

$$x_2 = 4$$

$$x_3 = 6$$

$$x_4 = \dots$$

Sequence

Definition

A *sequence* of real numbers is a function, u , mapping $\mathbb{N}$ to $\mathbb{R}$

- We will be dealing with infinite sequences
- Written $u(1), u(2), u(3), \dots$, or $(u_1, u_2, u_3, \dots)$
- The sequence is written $\{u_n\}$
- For example:

$$u_n = \frac{3n^2}{n^3 + 9}$$

gives the sequence

$$\frac{3}{10}, \frac{12}{17}, \frac{48}{73}, \dots$$

Definition

A *sequence* of a set X is a function, u , mapping $\mathbb{N}$ to X

- Ideal classes of sequences:
 - Arithmetic sequences
 - Geometric sequences
- Defining characteristics of sequences
 - Recurrent behavior: oscillation, convergence, divergence
 - Specific kinds of convergence.

Arithmetic Sequences

Definition

An arithmetic sequence is a sequence $\{u_n\}$ in which

$$u_{n+1} - u_n$$

is constant for all n

- Equivalent to saying that for some numbers a, d , the sequence looks like

$$a, a + d, a + 2d, a + 3d, \dots$$

or

$$u_n = a + (n - 1)d$$

- Is this sequence arithmetic? (If so, what are a, d ?)

$$u_n = 14 - 5n$$

Geometric Sequences

Definition

Geometric sequences: Each term is obtained by multiplying previous term by same number.

- Equivalent to saying that for some numbers a, x the sequence looks like

$$a, ax, ax^2, ax^3, \dots$$

or

$$u_n = ax^{n-1}$$

- What are a, x for this geometric sequence?

$$u_n = 5^{-n}$$

Limits of Sequence: Motivation

- Most sequences we will study will be neither arithmetic or geometric.
- How do we measure the behavior of sequences that change over time?
- Many sequences behave similarly as n gets large.
- One way to characterize these sequences is to identify where they are converging to, a limit.
- However, we need an objective criteria for determining whether two sequences are going to the same place.

Sequence Applications

- We saw that a sequence is some function $u(n)$ where n is an integer.
- One important such function is:

$$u(1) = \frac{X_1}{1}$$

$$u(2) = \frac{X_1 + X_2}{2}$$

$$u(3) = \frac{X_1 + X_2 + X_3}{3}$$

$$u(n) = \frac{X_1 + \dots + X_n}{n}$$

- The sample average with increasing sample size will converge (in probability) to the true mean.

Limits of sequences

Definition

A sequence $\{u_n\}$ tends to the limit z if for any positive real number ϵ there exists a positive integer, N , such that:

$$|u_n - z| < \epsilon \text{ for all } n > N.$$

Also written as

$$\lim u_n = z$$

$$\lim_{n \rightarrow \infty} u_n = z$$

$$u_n \rightarrow z \text{ as } n \rightarrow \infty$$

Limits of Sequences

■ Some Remarks

- If a sequence converges, it converges to one number, say z .
- We check by picking an $\epsilon > 0$, which is an arbitrary real-valued number as the maximum deviation from the proposed limit z .
- Then we find the N , which will depend upon ϵ , where the sequence is within the ϵ region of z .

Limits of Sequences

Theorem

$\left\{\frac{1}{n}\right\}$ converges to 0

Proof.

Without loss of generality (WLOG) select an ϵ . We need 0 to be within ϵ of N_ϵ :

$$\begin{aligned} \left| \frac{1}{N_\epsilon} - 0 \right| &< \epsilon \\ \frac{1}{N_\epsilon} &< \epsilon \\ \frac{1}{\epsilon} &< N_\epsilon \end{aligned}$$

For each ϵ , then, any $n > N_\epsilon > \frac{1}{\epsilon}$ will suffice. □

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Cauchy, (pronounced KohShee) Sequence: Motivation

In previous example, we could show that $\frac{1}{n}$ converges to 0 because we know its limit in advance. Can we show that a sequence has limit without knowing the limit *ex ante*?

A popular test for the existence of a limit is to determine if that sequence is Cauchy.

Divergent Sequence

Definition

A sequence that tends to a limit is called convergent; a sequence that doesn't is called divergent.

- A sequence that tends to infinity (or negative infinity) is divergent, but a sequence doesn't have to tend to infinity to be divergent. (What's an example of a divergent sequence that doesn't tend to infinity?)

Informal Definition of Cauchy Sequence

A sequence $\{x_n\}$ is Cauchy if it eventually stops bouncing around more than ϵ :

- Find two points in the sequence x_m and x_n .
- Take their difference. Is it getting smaller as m and n get big?

Formal Definition of Cauchy Sequence

Definition

*A sequence $\{u_n\}$ tends to the limit z if $\forall \epsilon > 0, \exists N \in \mathbb{N} :$
 $|u_n - z| < \epsilon, \forall n > N.$*

Definition

*$\{x_n\}$ is Cauchy if $\forall \epsilon > 0, \exists N \in \mathbb{N} :$
 $|x_m - x_n| < \epsilon$ when $m, n > N$*

Theorem

In $\mathbb{R}$, $\{x_n\}$ is convergent if and only if it's Cauchy.

Apply Cauchy Criterion

We can show the sequence defined by $x_1 = 1, x_2 = 2, \dots, x_n = n$ is not convergent:

- $|x_n - x_m| = |n - m| \geq 1$ for all n, m .
- Thus, for $\epsilon = 1$ there is no N such that $|x_n - x_m| < \epsilon$ whenever $n, m > N$.
- Therefore $\{x_n\}$ is not Cauchy, and does not tend to a limit.

Working with Limits

- Fact 1: For a real number x , $\lim_{n \rightarrow \infty} x^n = 0$ if and only if $-1 < x < 1$.
- Fact 2: For a real number b , $\lim_{n \rightarrow \infty} n^b = 0$ if and only if $b < 0$.

Working with Limits: Evaluate Dominating Term

What is the limit of

$$u_n = \frac{(-1)^n(n+1)^2}{n^4}$$

Hint: We can start by expanding the numerator.

$$u_n = \frac{(-1)^n(n^2 + 2n + 1)}{n^4}$$

$$u_n = \frac{(-1)^n(n^2)}{n^4} + \frac{(-1)^n(2n)}{n^4} + \frac{(-1)^n}{n^4}$$

What would happen if we replaced $(n+1)^2$ with $(n+1)^4$?
Don't do whole calculation!

Working with Limits: Evaluate Dominating Term

Another practice, though easier one, as $n \rightarrow \infty$, what's the limit of

$$u_n = \frac{n+1}{2n}$$

$$u_n = \frac{n}{2n} + \frac{1}{2n}$$

$$u_n = \frac{1}{2} + 0$$

Series

Definition

A *series* is sequence constructed from a sum of sequences.

- The series defined by a sequence $\{u_n\}$ is the sequence $\{s_n\}$ with

$$s_1 = u_1, s_2 = u_1 + u_2, s_3 = u_1 + u_2 + u_3, \dots$$

and more generally

$$s_n = \sum_{r=1}^n u_r$$

- Thus, series is a special type of sequence, it sums the terms in the sequence.

$\sum$ notation for series

$$s_n = \sum_{r=j}^n u_r$$

Here j is the start of the series, n is the end. We work through each number from j to n , plugging into r .

Suppose $j = 0$, $n = 10$ and $u_r = 2 * r$

$$\begin{aligned} s_n &= \sum_{r=0}^{10} u_r = 0 + 2 + 4 + 6 + 8 + 10 + 12 + 14 + 16 + 18 + 20 \\ &= \sum_{r=0}^3 u_r + \sum_{r=4}^7 u_r + \sum_{r=8}^{10} u_r \end{aligned}$$

Formula for Summing Algebraic Sequence

We are interested in the sum of first n terms in arithmetic sequence. Here is the derivation:

First, the natural numbers

$$s_n = 1 + 2 + \dots + (n - 1) + n$$

$$s_n = n + (n - 1) + \dots + 2 + 1$$

Adding these together we get

$$2s_n = (1 + n) + (1 + n) + \dots + (1 + n) = n(1 + n)$$

The first n terms of the natural numbers sum to

$$s_n = \frac{1}{2}n(1 + n)$$

Formula for Summing Algebraic Sequence

Recall that a general arithmetic sequence is $u_n = a + (n - 1)d$

We can use the same trick to find the sum of the first n terms of the sequence:

$$u_n = a + (a + d) + (a + 2d) + \dots + (a + (n - 2)d) + (a + (n - 1)d)$$

$$u_n = (a + (n - 1)d) + (a + (n - 2)d) + \dots + (a + 2d) + (a + d) + a$$

$$2u_n = n(2a + (n - 1)d)$$

$$2u_n = 2an + n(n - 1)d$$

$$u_n = an + n\left(\frac{(n - 1)d}{2}\right)$$

Summing Geometric Sequences

These same tricks can be extended to the geometric sequence:

$$s_n = a + ax + ax^2 + \dots + ax^{n-1}$$

Our strategy is to construct a forward and backward shifted sequence and then take differences to eliminate the middle.

Multiplying by x we get our original sequence shifted forward.

$$xs_n = ax + ax^2 + \dots + ax^n$$

We then define a sequence for all but the first term.

$$t = ax + ax^2 + \dots + ax^{n-1}$$

Then $s_n = a + t$ and $xs_n = t + ax^n$, we can eliminate t .

$$\begin{aligned} s_n - xs_n &= a - ax^n \\ s_n(1 - x) &= a(1 - x^n) \\ s_n &= a \left(\frac{1 - x^n}{1 - x} \right) \end{aligned}$$

An Example for Summing Geometric Sequence

Now suppose you save \$1000 in Citibank this year. Every future year, you save \$1000. The annual (compound) interest rate is 5 percent.

Then, a natural question will be: at year n , how much money do you make?

For the money you saved at year 0, you will get: $1000 * (1 + 0.05)^n$;

For the money you saved at year 1, you will get: $1000 * (1 + 0.05)^{n-1}$;

For the money you saved at year t , you will get: $1000 * (1 + 0.05)^{n-t}$;

Sum them together and you will have your total earnings:

$$1000 * (1 + 0.05)^n + 1000 * (1 + 0.05)^{n-1} + \dots + 1000 * (1 + 0.05)^1 = 13,206.79$$

Utility Application of Geometric Series

- Geometric series are often used to study the value of the future:
- $U(a) = a + a\delta + a\delta^2 + \dots + a\delta^{n-1}$
- δ is called the discount factor, and $0 < \delta < 1$
- We can use the properties of series to describe the long run (infinite) value of a choice.

Convergence of Geometric Series

For a geometric series,

$$s_n = a \left(\frac{1 - x^n}{1 - x} \right)$$

We know that $x^n \rightarrow 0$ as $n \rightarrow \infty$ if and only if $|x| < 1$

If $|x| < 1$ then $\lim_{n \rightarrow \infty} s_n = a \left(\frac{1-0}{1-x} \right) = \frac{a}{1-x}$

Applying this result to the long run value of a : $U(a) = \frac{a}{1-\delta}$.

$$\sum_{t=0}^{\infty} \delta^t u(x) = \frac{u(x)}{1-\delta}$$

Divergence of Geometric Series

Recall that $s_n = \sum_{k=1}^n k_n$.

Fact (*term test* for any series): If $\lim_{n \rightarrow \infty} k_n \neq 0$ then $\sum_{n=1}^{\infty} k_n$ diverges

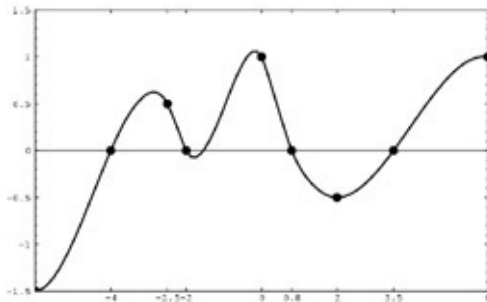
For a geometric series, $k_n = ax^{n-1}$

We know that $x^n \rightarrow 0$ as $n \rightarrow \infty$ if and only if $|x| < 1$

Therefore, a geometric series diverges whenever $|x| \geq 1$

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Limits of Functions: Motivation



We care about the limits of function because we care about how functions behave at a point.

It turns out that function can often only be described in the limit (recall probability example).

Limits of Functions

Definition

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$. We say that f has a limit L at x_0 if, for each $\epsilon > 0$, there is a $\delta > 0$ such that $|x - x_0| < \delta$ implies that $|f(x) - L| < \epsilon$.

- Limits are about the behavior of functions at points. Here x_0 .
- As with sequences, we let ϵ define an error rate
- δ defines an area around x_0 where $f(x)$ is going to be within our error rate

Limits of Functions: Example

Theorem

The function $f(x) = \frac{x^2-1}{x-1}$ has a limit of 2 at $x_0 = 1$.

Proof step 1, simplify

For all $x \neq 1$,

$$\begin{aligned} f(x) &= \frac{x^2 - 1}{x - 1} \\ &= \frac{(x + 1)(x - 1)}{x - 1} \\ &= x + 1 \end{aligned}$$

Limits of Functions: Example Continued

Show $f(x) = \frac{x^2-1}{x-1} = x+1$ has a limit of 2 at $x_0 = 1$.

Proof step 2:

Choose any $\epsilon > 0$ and set $x_0 = 1$. We're looking for δ_ϵ (a function of ϵ) such that

$$|x - 1| < \delta_\epsilon \quad \text{implies} \quad |(x + 1) - 2| < \epsilon$$

We guess a good δ_ϵ and use a direct proof:

Assume $\delta_\epsilon = \epsilon$,

$$|x - 1| < \delta_\epsilon$$

$$|x - 1| < \epsilon$$

$$|x + (1 - 1) - 1| < \epsilon$$

$$|(x + 1) - 2| < \epsilon$$

Remarks on Limits of Functions

$\lim_{x \rightarrow x_0} f(x)$ might or might not equal $f(x_0)$

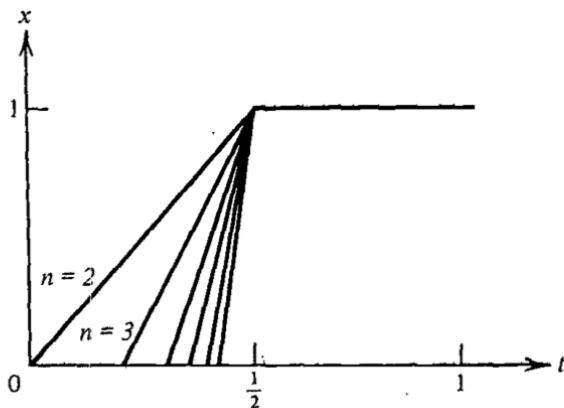
- It could be *different than* $f(x_0)$

$$f(x) = \begin{cases} +1 & \text{if } x \neq 4 \\ -1 & \text{if } x = 4 \end{cases}$$

- It could be *undefined*

$$f(x) = \begin{cases} +1 & \text{if } x < 0 \\ -1 & \text{if } x \geq 0 \end{cases}$$

Example of Cauchy sequence



■ Suppose
$$x_n(t) = \begin{cases} 0 & \text{for } 0 \leq t \leq \frac{1}{2} - \frac{1}{n} \\ nt - \frac{n}{2} + 1 & \text{for } \frac{1}{2} - \frac{1}{n} \leq t \leq \frac{1}{2} \\ 1 & \text{for } t \geq \frac{1}{2} \end{cases}$$

Algebra of Limits

Theorem

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ with limits A and B at x_0 . Then,

$$i.) \lim_{x \rightarrow x_0} (f(x) + g(x)) = \lim_{x \rightarrow x_0} f(x) + \lim_{x \rightarrow x_0} g(x) = A + B$$

$$ii.) \lim_{x \rightarrow x_0} f(x)g(x) = \lim_{x \rightarrow x_0} f(x) \lim_{x \rightarrow x_0} g(x) = AB$$

Suppose $g(x) \neq 0$ for all $x \in \mathbb{R}$ and $B \neq 0$ then $\frac{f(x)}{g(x)}$ has a limit at x_0 and

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow x_0} f(x)}{\lim_{x \rightarrow x_0} g(x)} = \frac{A}{B}$$

Power Series

- The following is the most important function in mathematics:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = \lim_{n \rightarrow \infty} \left(\frac{1}{0!} + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right)$$

- It has special properties when we get to derivatives.

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Continuity

Definition

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ and consider $x_0 \in \mathbb{R}$. We will say f is continuous at x_0 if for each $\epsilon > 0$ there is a $\delta > 0$ such that if,

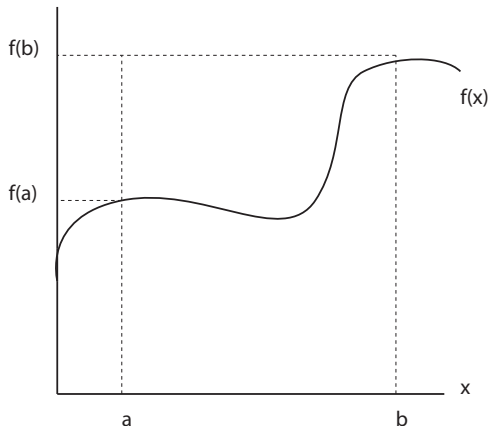
$$\begin{aligned} |x - x_0| &< \delta \text{ for all } x \in \mathbb{R} \text{ then} \\ |f(x) - f(x_0)| &< \epsilon \end{aligned}$$

- Previously $f(x_0)$ was replaced with L .
- Now: $f(x)$ has to converge on itself at x_0 .
- Continuity is more restrictive than limit.

The intermediate value theorem

Consider an interval $[a, b]$ in the real numbers.

If the function $f(x)$ is continuous for all $a \leq x \leq b$ then f takes every value between $f(a)$ and $f(b)$.



Proof outline of intermediate value theorem

- If we have any u between two values of the function, $f(a)$ and $f(b)$, we can find a $f(c) = u$
- Call S the set of numbers $x \in [a, b]$ such that $f(x) \leq u$. We want to show that the function takes on the value $f(c) = u$ at some $c \in S$. We will show that this occurs at the supremum (the lowest number greater than or equal to every element of S).
- Pick a small area around $f(x)$, use properties of continuity of f and the real numbers, find that the only value that can satisfy these conditions is $f(c) = u$